Smart Online Examination Registration systems

a) Input conditions and equivalence classes

Course : Registered / Not Registered

Attendance : valid 0-100% ; invalid below 0% or above 100%.

Fee : paid / unpaid

Register Number: valid exactly 12 digits ; otherwise invalid

b) ECP and BVA

Attendance	Expected output
74%	Reject
75%	Accept if other condition are valid
76%	Accept if other condition are valid

c) Decision Table

course	Attendance	Fee	Register Number	Result
valid	valid	paid	Valid	Accept
Invalid	Valid	Paid	Valid	Reject
Valid	Invalid	Paid	Valid	Reject
Valid	Valid	unpaid	Valid	Request payment
Valid	Valid	Paid	Invalid	Display Error

d) Random Testing

Test Random attendance values (42%, 88%, 103%) and Register Numbers (12 digits, 11 digits, letters). Random testing may reveal number unexpected defects missed by planned Tests.

e) Test adequacy

priortize high-risk conditions, test valid / Invalid/ boundary inputs, measurement and branch coverage and remove redundant tests.

2) Online Banking Application

a) state transition diagram

Logged out

Valid credentials + OTP ↓

Logged In

Valid beneficiary + amount ↓

Transfer in progress

success → Transfer successful

Network failure → Transfer failed

3 invalid OTPs → Account Locked

Authorized recovery → Access restored

b) OTP boundary Testing

Test	Input	Expected
1	correct OTP, 1st attempt	Login success
2	Two wrong, then correct	Login success
3	Three wrong OTPs	Account Locked

c) pairwise Testing

Test	Device	Browser	Network
1	Laptop	chrome	Wi-Fi
2	Laptop	Edge	Mobile Data
3	Mobile	chrome	Mobile Data
4	Mobile	Edge	Wi-Fi

d) use-case and exploratory testing

Duplicate transfer: Ensure money is not deducted twice

session expiry: Require Reauthentication

Network failure: Verify final transaction status

Wrong beneficiary: Verify recipient details before transfer

e) COTS gateway testing

Test integration, browser / device compatability timeouts and security. Source code unavailability limits internal testing. use vendor documentation sandbox testing and Logs.

3) Smart ATM Fund- Transfer system

a) CFG

START -> amount <= 0 ?

yes -> invalid amount -> STOP

NO -> amount > Balance ?

yes -> insufficient Balance -> STOP

No -> amount > dailylimit ?

Yes -> Daily Limit Exceeded -> STOP

No -> Deduct amount -> Transfer successful -> STOP

b) cyclomatic complexity

There are 3 decision nodes

$V(G) = D + 1 = 3 + 1 = 4$

c) Test cases

Assume Balance = ₹ 10,000 and daily Limit = 25,000

=> 0 - invalid amount

12,000 - insufficient Balance

6,000 - Daily limit exceeded

3,000 - success ; Balance ₹ 7,000

d) coverage Types

Statement : Execute every statement

Branch : Test yes and No outcomes of each decision

condition : Make each condition True and False

path : Execute all four independent paths

e) Test adequacy

100% statement coverage does not guarantee defect-free software. Mutation testing changes conditions or remove checks to verify whether tests detect errors.

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4) Online Login security system

a) Loop and decision structures

The program uses a While loop (count < 3) and an IF condition to compare passwords. Correct password gives success ; three incorrect attempts lock the account

b) Loop - testing cases

password sequence	Expected
correct first	success
Wrong correct	success
Wrong, Wrong, correct	success only
Three Wrong	Account locked
Zero attempts	No login

c) Defect and correction

used a logged in boolean flag set it to true after successful login and display account Locked only when logged in == false and count == 3

d) coverage

condition and branch coverage

path coverage

Boundary Testing

e) Mutation Testing

change count < 3 to count < 2, remove Break; or make password comparison always false Test should detect the defects.

d) EOP and BVA

Students : Test 0, 1, 100, 101

Subjects : Test 0, 1, 10, 11

Marks : Test -1, 0, 1, 100, 101

e) Test adequacy

Measure statement, branch, loop, boundary, and data-flow coverage, priortize invalid inputs and zero-subject cases. Defects are disolved.